

```
module tb_peak;

reg clk, rst;

reg [7:0] data_in;

wire peak;

peak_detector uut (
    .clk(clk),
    .rst(rst),
    .data_in(data_in),
    .peak(peak)
);

// clock generation
always #5 clk = ~clk;

initial begin
    clk = 0;
    rst = 1;
    data_in = 0;

    #10 rst = 0;

    // sample stream
    data_in = 5; #10;
    data_in = 8; #10;
    data_in = 12; #10;
    data_in = 9; #10;
    data_in = 6; #10;
```

```
data_in = 10; #10;
```

```
data_in = 7; #10;
```

```
#50 $finish;
```

```
end
```

```
// waveform dump (for GTKWave)
```

```
initial begin
```

```
    $dumpfile("dump.vcd");
```

```
    $dumpvars(0, tb_peak);
```

```
end
```

```
// clean printing every clock
```

```
always @(posedge clk) begin
```

```
    $display("Time=%0t | din=%0d | peak=%b", $time, data_in, peak);
```

```
end
```

```
endmodule
```